
Model Parameter Reference

Every parameter in the decussated-lattice models:
measured, derived, or constant — and how they relate

Cameron Renteria

NIH–NIDCR Postdoctoral Fellow, University of Washington (MSE + OHS)

`biomimetic_pipeline` — continuous-twist parametric route

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Scope

This reference documents the **continuous_twist** parametric lattice (`lattice.cad.py` via `feature_to_cad.map_morphometrics`). Each parameter is classified **measured** (traceable to a measured SOM morphometric), **derived** (computed from other CAD parameters), or **design const** (a documented design or numerical constant, not a measurement). **gap** flags a traceability gap noted in the audit. The zero-free-parameter claim rests on the **measured** rows; the **design const** and **gap** rows must be disclosed as such.

1 Master ledger

Every emitted parameter at a glance. Canonical values are for the biomimetic `live_001` specimen at `biology_scale_factor=500`.

Parameter	Class	Value	Source / formula
ROD_DIAMETER	measured	2.1280 mm	$\text{mean}(\text{rod_diameter_um}) \times \text{scale} \times 10^{-3}$
CENTER_SPACING	measured	3.1919 mm	<code>band_width_um</code> , clamped to ceiling $1.5 \times \text{ROD}$
RING_ROTATION	measured	$\{+2.27 \dots -54.17\}^\circ$	$\int \text{pitch_signed_deg}$ over depth, anchored to <code>band[0]</code>
ROD_TAPER_FACTOR	measured	0.0	measured -0.041 , clamped to $[0, 1]$ (logged)
N_BRIDGE_LAYERS	measured/sweep	4	periodicity (degenerate $\rightarrow 2$) <i>or</i> sweep-injected
<code>per_ring_diameter</code>	measured (6 vals)		graded rod profile — dropped at consume
<code>rod_cross_section</code>	measured gap	$b = 3.62$	from eccentricity — dropped (built circular)
<code>twist_perturbation_amplitude_deg</code>	measured gap	15	$(\text{tortuosity} - 1) \times 180 = 20.69$, saturated at 15, clamp unlogged
BRIDGE_DIAMETER	derived	1.7024 mm	<code>bridge_ratio</code> $\times \text{ROD} = 0.80 \times \text{ROD}$
BRIDGE_Z_OFFSETS	derived	N z-levels	safe band of (ENAMEL, PLATE_OVERLAP, BRIDGE, jitter)
<code>bridge_ratio</code>	design const gap	0.80	hand-tuned; sole BRIDGE determinant; not in morphometrics
<code>biology_scale_factor</code>	design const	500	$\mu\text{m} \rightarrow \text{mm}$ print-scale convention
N_RINGS	design const	5	hardcoded (dead inference code)
ENAMEL_THICKNESS	design const	20 mm	stock default (no measured source)
OCCT_TANGENT_JITTER_MM	design const	0.50	numerical OCCT-escape; per-trial; mixed across sweep
PLATE_OVERLAP/THICKNESS	design const	0.5 / 1.2	stock plate geometry
Z_SAMPLES	design const	50	helical sweep resolution (affects OCCT tangent)
MATERIAL_E	design const	3000 MPa	SLA resin; recorded in metrics
MATERIAL_NU	design const	0.40	SLA resin Poisson ratio
<code>mesh_size_mm</code>	design const gap	0.5	$\sim 3\text{--}4$ elem/cross-section; no convergence study

Parameter	Class	Value	Source / formula
vm_target_mpa	design const	200	design stress target (the one objective field consumed)
seed_disp_mm / solver_tolerance_pct	design const gap	1.0 / 5%	dead YAML — strain_solver defaults coincide
COMPRESS_DISP_MM	derived	solved	bisected until avg von Mises = vm_target_mpa

2 Measured-derived parameters (the zero-free-parameter core)

These trace to the SOM morphometrics; they are the architecture the manuscript claims is inherited from measurement.

ROD_DIAMETER = 2.127 955 mm.

$\text{mean}(\text{depth_profiles.rod_diameter_um_mean}) \times \text{biology_scale_factor}(500) \times 10^{-3}$. The measured axial profile is collapsed to a scalar (the radial grading is recovered in `per_ring_diameter`, then dropped — see gaps).

CENTER_SPACING = 3.191 932 mm.

Derived from `band_width_um` then clamped; the operative bound is the ceiling $1.5 \times \text{ROD}$. Floor $\max(1.05 \times \text{ROD}, 2 \times \text{SLA_min})$ guarantees rods cannot overlap.

RING_ROTATION (**per ring** $0 \dots N_{\text{RINGS}}$).

Cumulative integral of `depth_profiles.pitch_signed_deg` over depth, sampled at i/N_{RINGS} , then the whole dict is anchored so ring 0 = `bands[0].mean_direction`. Canonical: $\{0: +2.27, 1: -5.84, 2: -15.11, 3: -31.00, 4: -44.95, 5: -54.17\}^\circ$. *Modeling caveat:* pitch (an angle) is integrated over depth and used directly as a twist increment — an implicit “local pitch $\approx$ twist per depth” assumption to state in Methods.

ROD_TAPER_FACTOR = **0.0**.

Measured taper -0.041 (slight narrowing) clamped to $[0, 1]$; the clamp *is* logged in `provenance.scale_clamps`.

N_BRIDGE_LAYERS.

The measured derivation from `dominant_wavelength_um_mean(85)` is degenerate (rounds to 0, floor-rescued to 2). In the canonical sweep it is the *design variable* (4–9), injected via `extra_overrides` — legitimately swept, but flag it as injected, not measured.

3 Derived (geometric) parameters

BRIDGE_DIAMETER = 1.702 364 mm = $0.80 \times \text{ROD_DIAMETER}$.

Tied to ROD via `bridge_ratio`; must stay strictly $< \text{ROD}$ (lattice_cad invariant).

Fixed bug — the reason for this audit

Overriding ROD_DIAMETER used to *silently re-derive* this value ($0.80 \times \text{thick rod} = 2.47 \text{ mm}$), shipping the whole thick FEA arm on wrong geometry. As of 2026-05-31 `map_morphometrics` honors an explicit BRIDGE_DIAMETER/CENTER_SPACING override and *raises* on a lone ROD override. Locked by `tests/test_value_carrythrough.py`.

BRIDGE_Z_OFFSETS (N z-levels).

`compute_safe_bridge_elevations`: N uniformly-spaced elevations in a safe band $[\text{plate_overlap} + m, \text{cut_top} - \text{plate_overlap} - m]$ where $m = \frac{1}{2}\text{BRIDGE} + \text{clearance}$, then shifted up by `OCCT_TANGENT_JITTER_MM` (top pinned). Depends on `ENAMEL`, `PLATE_OVERLAP`, `BRIDGE`, jitter, N — not on a measurement.

COMPRESS_DISP_MM.

Not an input: the strain solver bisects displacement until the average von Mises stress equals `vm_target_mpa` (200 MPa), so all designs compare at matched representative stress.

4 Design constants (documented, *not* measured)

These must be disclosed as conventions/constants in the manuscript — they are not free fits, but neither are they measured.

- `bridge_ratio=0.80` — the **sole** determinant of every bridge diameter, hand-tuned, in neither morphometrics file. The most load-bearing non-measured constant. **gap**
- `biology_scale_factor=500` — the $\mu\text{m} \rightarrow \text{mm}$ print-scale convention (reverse-engineered to the stock ~ 2 mm rod). Correctly applied and logged.
- `N_RINGS=5` — hardcoded; the inference routine ignores its arguments (dead code). Sizes `RING_ROTATION`, `per_ring_diameter`, and the twist grid. **gap**
- `ENAMEL_THICKNESS=20` mm, `PLATE_OVERLAP=0.5`, `PLATE_THICKNESS=1.2`, `Z_SAMPLES=50`, `FILLET/CHAMFER=0`, `JUNCTION_SPHERE_FACTOR=0` — stock geometry/meshing constants.
- `OCCT_TANGENT_JITTER_MM=0.50` — a *numerical* sub-mm bridge-height nudge to dodge OCCT boolean tangency. Mixed across the sweep (0.0/0.15/0.50), recorded per-trial in each `cad_params.json`; not a design quantity. **gap** (not in provenance summary).

5 FEA inputs

- `MATERIAL_E=3000` MPa, `MATERIAL_NU=0.40` — SLA resin. *Verified* required-and-passed: `run_one_iteration` demands them, sets them in the solver env, and `metrics.json` records `material_E_MPa/material_nu`. (The audit’s “silent 85000 MPa fallback” was over-stated — there is no path that reaches the stock default.) Cite the resin datasheet in Methods.
- `vm_target_mpa=200` — the design operating point; the one objective-config field actually consumed.
- `mesh_size_mm=0.5` — ~ 3 –4 elements across a rod/bridge cross-section on a geometry-sensitive model; no convergence study in-repo. **gap**
- `seed_disp_mm=1.0`, `solver_tolerance_pct=5%` — declared in `crack_deflection.yaml` but **not forwarded** to the solver; `strain_solver` uses its own equal defaults, so editing the YAML changes nothing. **gap** (dead config / false provenance).

6 How the parameters relate

The coupling web

Rod $\rightarrow$ bridge $\rightarrow$ spacing (the invariant chain). `BRIDGE_DIAMETER` = $0.80 \text{ ROD_DIAMETER}$ and must stay $< \text{ROD}$; `CENTER_SPACING` = 1.5 ROD_DIAMETER (ceiling) and must stay $> 1.05 \text{ ROD}$. So `ROD` sets both bridge thickness and ring pitch.

Thick-rod variant. `ROD_DIAMETER` = `CENTER_SPACING` - 0.025 (0.025 mm worst-case rod-surface gap), with `BRIDGE_DIAMETER` held at the biomimetic 1.702 mm. Generated via `make_thick_rod_variant.py`, which sets `ROD` on the param dict directly — the only correct path, since `map_morphometrics` would otherwise re-derive `BRIDGE/CS` from the grown rod.

Bridge placement. `BRIDGE_Z_OFFSETS` span = $f(\text{ENAMEL_THICKNESS}, \text{PLATE_OVERLAP}, \text{BRIDGE_DIAMETER}, \text{OCCT_TANGENT_JITTER_MM})$ the N layers are uniform within it. Denser $N \Rightarrow$ tighter vertical gap $\Rightarrow$ more OCCT tangency risk (hence the jitter).

Discretization. `N_RINGS`=5 sizes `RING_ROTATION`, `per_ring_diameter`, and the twist sampling grid. `Z_SAMPLES`=50 sets the helical sweep resolution.

FEA closure. `COMPRESS_DISP_MM` is bisected until $\langle \sigma_{vM} \rangle = \text{vm_target_mpa}$ given `MATERIAL_E`; all metrics are read at that matched stress.

7 Known gaps & disclosures (from the audit)

1. **Dropped measured geometry (deferred decision).** `per_ring_diameter` (radial rod grading) and `rod_cross_section` (ellipticity $a/b=3.62$) are *emitted* but not consumed by the stock CAD — every rod is built at the scalar `ROD_DIAMETER` and as a circle. The decussation architecture (twist, bands, bridges) is measured-derived; the per-rod cross-section is simplified. Decision on disclose-vs-implement is deferred.
2. **Untraceable constants.** `bridge_ratio` and `N_RINGS` should be derived from a measured quantity or explicitly labeled design constants in provenance.
3. **Dead solver config.** `seed_disp_mm/solver_tolerance_pct` should be wired through or removed; do not echo unconsumed values as provenance.
4. **Unlogged saturating clamp.** `twist_perturbation_amplitude_deg` is fully set by the 15° cap (measured 20.69°); log the saturation.
5. **Jitter provenance.** Record `OCCT_TANGENT_JITTER_MM` per run so committed trials are re-derivable from source.
6. **Fixed.** The `BRIDGE_DIAMETER/CENTER_SPACING` silent re-derivation on override (the bug that shipped inflated thick bridges) is fixed and regression-tested.

Bottom line

The lattice *architecture* — rod diameter, ring spacing, per-ring decussation twist, taper, and the swept bridge-layer count — is measured-derived and now value-locked by tests. The bridge *thickness* (via `bridge_ratio`), the $\mu\text{m} \rightarrow \text{mm}$ `biology_scale_factor`, ring count, part height, material, and mesh are documented constants. Two measured cross-section details (grading, ellipticity) are currently simplified away. Disclosing exactly this split is what makes the zero-free-parameter claim honest.